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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2015-2016

CE2005/CZ2005/CSC205 – OPERATING SYSTEMS

Nov/Dec 2015 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 6 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. All questions carry equal marks.

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1. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) The read/write instructions to hard disks are privileged instructions.

(2 marks)

(ii) If a system has a process state transition from running to ready, the system must use a preemptive scheduling algorithm.

(2 marks)

(iii) Compared with message passing, shared memory is simpler and more efficient.

(2 marks)

(iv) Multi-programmed systems are designed to have the capability of running processes on multiple CPUs.

(2 marks)

(v) To improve the CPU utilization, a scheduling algorithm should always give high priorities to jobs with intensive CPU computation, rather than jobs with intensive I/O.

(2 marks)

Note: Question No. 1 continues on Page 2.

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(b) Consider the following set of processes in Table Q1, with the length of the CPU-burst time given in milliseconds (ms). A smaller priority number means a higher priority. The processes are assumed to have arrived in the order of P1 and P4 at time 0 ms, P2 at time 2 ms, P5 at time 3 ms, and P3 at time 4 ms. Give their average waiting times when the system uses each of the following algorithms: Non-preemptive Shortest Job First scheduling and Preemptive Priority based scheduling.

Table Q1

Process	Burst Time	Arrival Time	Priority
P1	2	0	1
P2	3	2	2
P3	1	4	3
P4	2	0	3
P5	3	3	1

(6 marks)

(c) Jim has written a C/C++ program, as shown in Figure Q1. Explain the role of the operating system when Lines 3-5 are executed.

```
        void main () {
Line 1:    int range=201;
Line 2:    int buffer [200];
Line 3:    for(int i=0; i<=range; i++ ) {
Line 4:      buffer[i]=i;
Line 5:    }
Line 6: }
```

Figure Q1

(3 marks)

(d) Jim designs a variant of the round robin scheduling algorithm where the entries in the ready queue are pointers to PCBs (Process Control Blocks). For some processes, N pointers to the same process are put in the ready queue (N>1). For other processes, only one pointer is put in the ready queue in his algorithm.

(i) What are the major advantages and disadvantages compared with the basic round robin algorithm?

(3 marks)

(ii) How would you modify the basic round robin algorithm to achieve the same effect without the duplicate pointers?

(3 marks)

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2. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) The correctness of a solution to the critical section problem should not depend on the speed of process execution.

(2 marks)

(ii) Since atomic instructions can solve the race condition problem, we should design all instructions in the system to be atomic.

(2 marks)

(iii) Real-time systems do not have deadlock problems.

(2 marks)

(iv) Deadlock will not occur in the Dining-Philosophers problem, if the system preempts a philosopher who cannot get another chopstick for a long time.

(2 marks)

(v) If a deadlock occurs in the system, the resource allocation graph must contain at least one cycle.

(2 marks)

(b) Consider a system with three processes (P, Q, and R) and one resource type. The available number of instances of the resource is 10. The Max resource requirements of P, Q and R are 8, 4 and 9, respectively. The processes issue the requests one by one (r1, r2, …, r6), as shown in Table Q2a. How does the system handle those requests according to Banker's algorithm?

Table Q2a

No.	Process	#instance requested
r1	R	2
r2	P	4
r3	Q	2
r4	P	2
r5	R	1
r6	Q	2

(5 marks)

Note: Question No. 2 continues on Page 4.

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(c) Suppose a system has three processes (R, M and P), with two shared bounded buffers (B1 and B2). Both B1 and B2 can only hold one data item. Process R receives data from an input device and stores the data to Buffer B1. Process M takes data from Buffer B1, processes the data and stores the result to Buffer B2. Process P takes the result from Buffer B2 and prints it with a printer. Use semaphores to implement the synchronization among R, M and P, which are shown in Table Q2b. You are only allowed to add semaphore definitions and semaphore operations (Wait and Signal) into the pseudo code for R, M and P.

Table Q2b

Process **R**	Process **M**	Process **P**

while (1) {
get one data item
from the device;
put the data to
B1;
}
	while (1) {
	get one data item
	from ***B1***;
	…
	process data
	item;
	…
	put the result to
	B2;
	}
		while (1) {
		get result from
		B2;
		print the result;
		}

(5 marks)

(d) A programmer has implemented the producer and consumer processes for the bounded buffer problem. The code is shown in Figure Q2. Determine all the possible scenarios when deadlock occurs, and propose a way to solve the deadlock problem.

Shared data declaration:
item buffer[BUFFER_SIZE];
semaphore *full=0, empty=n, mutex=1*;

Producer process:	**Consumer process:**
item *nextProduced*;	item *nextConsumed*;
while (1) {	while (1) {
…	*wait*(*mutex*);
produce *nextProduced;*	*wait*(*full*);
…	…
wait(*mutex*);	*nextConsumed* = an item from b*uffer;*
wait(*empty*);	…
…	*signal*(*mutex*);
add *nextProduced* to *buffer;*	*signal*(*empty*);
…	…
signal(*mutex*);	consume the item *nextConsumed;*
signal(*full*);	…
}	}

Figure Q2

(5 marks)

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3. (a) Indicate whether the following statements about memory management are true or false. Justify your answers.

(i) Load time address binding generates identical logical and physical addresses.

(2 marks)

(ii) External fragmentation may occur if segmentation is used for memory allocation.

(2 marks)

(iii) Using Translation Look-aside Buffer (TLB) will always reduce the effective memory access time.

(2 marks)

(iv) Code can be shared amongst processes in paging or segmentation only if it is reentrant.

(2 marks)

(v) Increasing the degree of multiprogramming will always increase CPU utilization.

(2 marks)

(b) Compared to the conventional, single-level page table scheme, what are the advantages and disadvantages of the inverted page table scheme?

(4 marks)

(c) Assume an operating system supports a 20-bit virtual address. The page size is 512 bytes and page table entry size is 4 bytes. To translate the logical address 654321 (in decimal number) to the corresponding physical address, which page table entry should be used in each of the following two cases?

- Page table in single-level paging
- Outer page table in two-level paging

(5 marks)

(d) Describe the procedure for page fault handling and identify the major overheads involved.

(6 marks)

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4. (a) Indicate whether the following statements about file system and I/O are true or false. Justify your answers.

(i) Using tree-structured directories, a file can have different names.

(2 marks)

(ii) Accessing a file using symbolic link takes longer time than using hard link.

(2 marks)

(iii) Assuming the protection bits of a directory are "r w x r - - r - -", all the users in this Unix-based system are able to list the content of this directory.

(2 marks)

(iv) All file allocation methods suffer from internal fragmentation.

(2 marks)

(v) Device drivers are part of the kernel I/O subsystem.

(2 marks)

(b) Assume a file has 10 data blocks (numbered 0 to 9) and the directory containing the file is already in the memory. To delete data block 5, how many disk reads and writes are required for each of the following file allocation methods?

- Contiguous allocation
- Linked allocation
- Indexed allocation (assuming index blocks are in disk)

(6 marks)

(c) Circular SCAN (C-SCAN) is a variation of SCAN disk scheduling algorithm. It is designed to provide a more uniform waiting time.

(i) Assume a disk drive has 200 cylinders, numbered 0 to 199. The following requests, arranged according to arrival order, are in the disk queue: 45, 125, 60, 95, 150, 20 and 185. The disk head is currently at cylinder 75, servicing a request, and its current moving direction is towards cylinder 0. To serve these requests in the disk queue, which disk scheduling algorithm, SCAN or C-SCAN, will result in less head movement?

(6 marks)

(ii) Explain why C-SCAN, compared with SCAN, reduces the maximum delay that may be experienced by a request.

(3 marks)

END OF PAPER

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1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.